Design of an 8-Button USB Game Controller Using KiCad and Onshape

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Abstract

This report presents the design of a compact eight-button USB game controller, done fully on the computer. The electronic circuit was drawn in KiCad: each of the eight buttons (Up, Down, Left, Right, A, B, X, Y) joins one input pin of an Arduino Pro Micro board to ground, so no external resistors are needed. The schematic passed the electrical rules check with 0 errors and 0 warnings. The enclosure was modelled in Onshape as a 150 mm by 80 mm case in two halves, with eight 12 mm holes for the button caps. The case halves, eight button caps and a board block were assembled with ten fastened mates, and the caps fit their holes. The design uses nine electronic parts, is easy to extend with a joystick or wireless module, and is ready to be 3D-printed and built.

Keywords: game controller, KiCad, Onshape, schematic, Arduino Pro Micro, enclosure design

1. Introduction

A game controller is a small product that mixes electronics and mechanical design. The circuit has to read every button correctly, and the case has to hold the board and put each button where a thumb can reach it. If the two are designed separately, small measurement mistakes show up only after parts are bought and built. Designing both on a computer first lets the team find and fix such mistakes at no cost.

This project designs a simple eight-button controller with a D-pad (Up, Down, Left, Right) and four action buttons (A, B, X, Y). The brain of the controller is an Arduino Pro Micro, a small board whose microcontroller supports USB directly, so it can appear to a PC as a gamepad. The circuit was drawn and checked in KiCad, a free electronics design program. The case and the button caps were modelled and assembled in Onshape, a 3D design program that runs in a web browser.

The objectives of the project are:

- to draw a correct and error-free circuit for an eight-button controller,
- to design a two-piece case with a hole for each button and a place for the board,
- to join all parts in one assembly and check that they fit together, and
- to keep the design simple, with few parts and no external resistors.

Section 2 describes the system and its parts. Section 3 explains the circuit, and section 4 explains the case. Section 5 gives the results of the checks, and section 6 concludes the report and lists possible improvements.

2. Proposed system

The controller has four blocks. The eight buttons are the inputs. The Arduino Pro Micro reads them. A USB cable carries the signals to the PC, and the PC sees the controller as a gamepad. A two-piece case holds the board and carries the buttons.

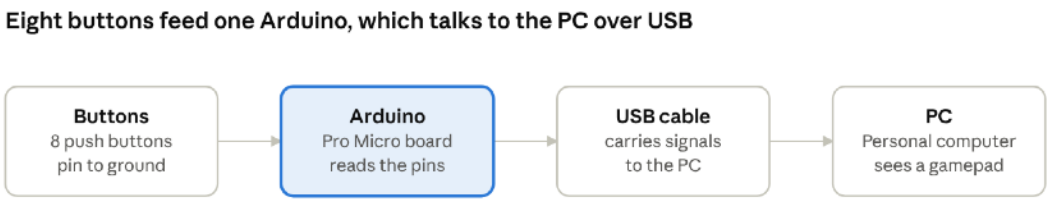


Figure 1. Block diagram of the controller

When no button is pressed, each input pin of the Arduino sits at a high level because of the board's internal pull-up resistor. When a button is pressed, it connects its pin to ground, and the pin reads low. The firmware on the Arduino (an outline is given in section 3.1) turns each low pin into the matching gamepad button.

Table 1 lists all parts of the controller.

Table 1. Parts of the controller

Part	Reference	Quantity	Purpose
Arduino Pro Micro	U1	1	Reads the buttons and talks to the PC as a gamepad over USB
Push buttons	SW1 to SW8	8	Up, Down, Left, Right, A, B, X, Y
USB cable	-	1	Connects the controller to the PC
Case, bottom half	-	1	Holds the board
Case, top half	-	1	Has the eight button holes
Button caps	-	8	The part the thumb presses

3. Circuit design in KiCad

The circuit was drawn on one A4 sheet in KiCad (version 10.0.7). Each button is a push button that has one pin wired to an input pin of the Arduino and the other pin wired to ground. This is the whole circuit, which is why it needs so few parts.

The basic symbol library has no Arduino Pro Micro, so a 9-pin connector, U1, stands in for the board. Connector pins 1 to 8 stand for Arduino pins D2 to D9, and pin 9 stands for GND. A ground symbol (GND) is attached to the free pin of every button and to pin 9. A power flag (PWR_FLAG) is placed on the ground net. It is only a marker that tells the checker that the ground is supplied, not a physical part.

No external resistors are needed. In the firmware each input pin is set to use the Arduino's built-in pull-up resistor, so a pin reads high until its button is pressed and low while it is pressed.

Table 2. Button wiring

Button	Reference	Arduino pin	Connector pin (U1)
Up	SW1	D2	1
Down	SW2	D3	2
Left	SW3	D4	3
Right	SW4	D5	4
A	SW5	D6	5
B	SW6	D7	6
X	SW7	D8	7
Y	SW8	D9	8
Ground (all buttons)	GND	GND	9

The finished schematic is shown in Figure 2. The title block carries the project title, the names of the team and the date.

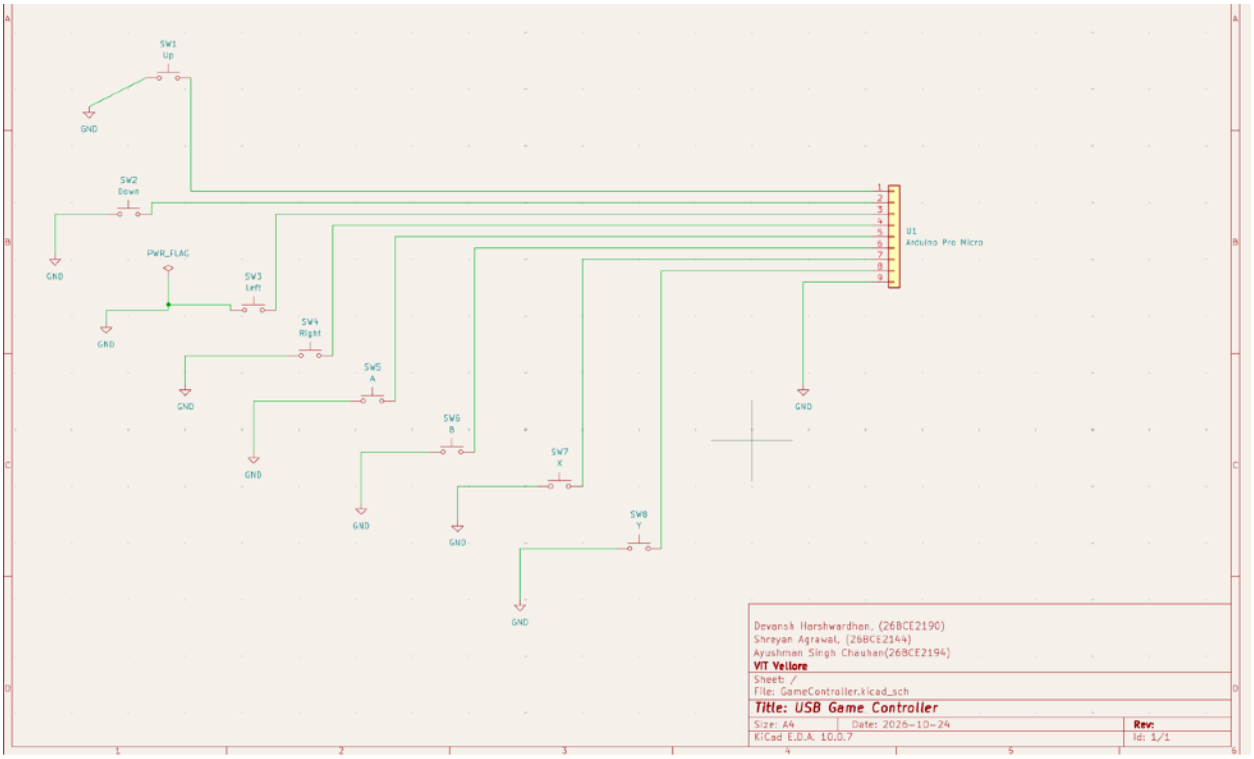


Figure 2. Schematic of the eight-button controller drawn in KiCad

3.1 Firmware (design only)

The controller needs a small program on the Arduino. The program sets pins D2 to D9 as inputs with the internal pull-up resistor, reads them in a loop, and reports each pressed button to the PC as a gamepad button. It uses the Arduino Joystick library by M. Heironimus, which works with boards built on the ATmega32U4 chip, such as the Pro Micro. The sketch below was written for this report. It has not been compiled or tested on a board.

```
#include <Joystick.h>

// Gamepad with 8 buttons and no axes
Joystick_ Joystick(JOYSTICK_DEFAULT_REPORT_ID, JOYSTICK_TYPE_GAMEPAD,
                   8, 0,
                   false, false, false, false, false, false,
                   false, false, false, false, false);

// Arduino pins in the order Up, Down, Left, Right, A, B, X, Y
const uint8_t pins[8] = {2, 3, 4, 5, 6, 7, 8, 9};

void setup() {
  for (uint8_t i = 0; i < 8; i++) {
    pinMode(pins[i], INPUT_PULLUP);   // pin reads HIGH until pressed
  }
  Joystick.begin();
}

void loop() {
  for (uint8_t i = 0; i < 8; i++) {
    bool pressed = (digitalRead(pins[i]) == LOW);
    Joystick.setButton(i, pressed);
  }
  delay(5);   // short pause so the buttons are not read too often
}
```

A button press connects its pin to ground, so the pin reads LOW. That is why the program treats LOW as pressed. The pin order in the program matches Table 2.

3.2 Electrical calculations

No external resistor is drawn on any button, because each pin uses the pull-up resistor inside the ATmega32U4 chip. The datasheet gives this resistor as roughly 20 kΩ to 50 kΩ. With a 5 V board, the current through one pressed button follows Ohm's law, I = V / R:

- with 50 kΩ: I = 5 V / 50 kΩ = 0.10 mA
- with 20 kΩ: I = 5 V / 20 kΩ = 0.25 mA

If all eight buttons are pressed at once, the worst case is 8 × 0.25 mA = 2 mA. This is far below the 20 mA that is recommended for one pin and far below the current a USB port provides (at least 100 mA before the device is configured, and up to 500 mA after). So the buttons do not stress the board or the USB port.

A push button can bounce for a few milliseconds when it is pressed, which means the contact opens and closes quickly several times. The 5 ms pause at the end of the loop in the program keeps these quick changes from causing false presses.

4. Mechanical design in Onshape

The case, the button caps and a block that stands for the Arduino board were modelled in Onshape, with all sizes in millimetres. The case is a rounded box in two halves, a bottom half that holds the board and a top half that carries the eight buttons. Each half is a hollow shell with a 2 mm wall.

Table 3. Main dimensions

Item	Size
Case outside size	150 mm long by 80 mm wide
Corner radius	10 mm
Wall thickness	2 mm
Bottom half height	12 mm
Top half height	13 mm
Button hole	12 mm diameter
Button cap, flange	14 mm diameter, 2 mm thick
Button cap, post	10 mm diameter, 8 mm above the flange
Arduino block	18 mm by 33 mm by 5 mm

The bottom half is open at the top, and the top half is open at the bottom, so together they close into one box. The eight holes are cut through the roof of the top half. The D-pad holes form a plus shape on the left, and the A, B, X, Y holes form a diamond on the right. The Arduino block lies flat between the two groups, so it does not sit under any hole. Table 4 gives the position of every hole, measured from the centre of the case.

Table 4. Hole positions

Button	x (mm)	y (mm)
Up	-40	20
Down	-40	-20
Left	-60	0
Right	-20	0
A	60	0
B	40	-20
X	40	20
Y	20	0

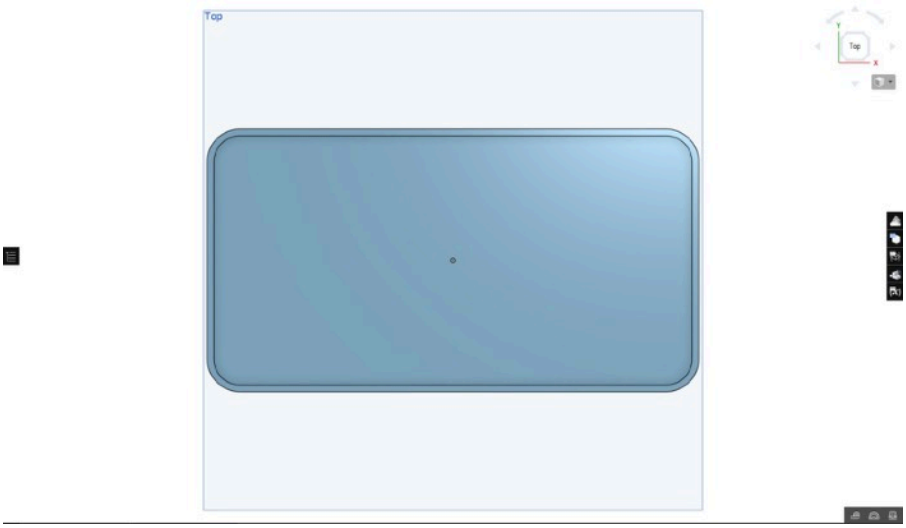


Figure 3. Bottom half of the case, seen from above

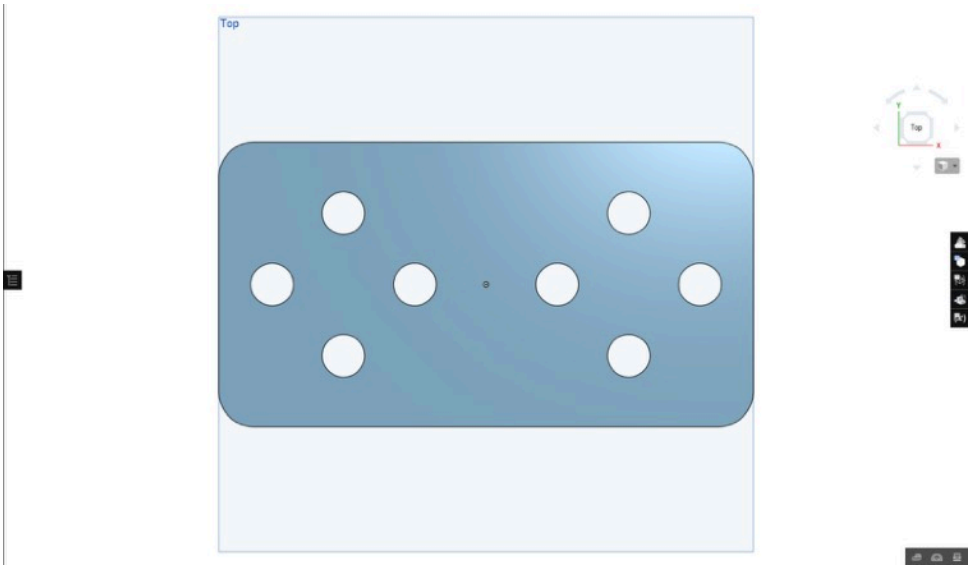


Figure 4. Top half of the case with the eight button holes

A button cap is a flat disc with a round post on top, like a mushroom. The post is 10 mm wide and passes through a 12 mm hole, so it has 1 mm of clearance on each side. The disc is 14 mm wide, which is wider than the hole, so the cap cannot fall out of the case.

All parts were joined in one Onshape assembly of eleven parts: the bottom half, the top half, the Arduino block and eight button caps. The bottom half is fixed, and the other ten parts are joined to it with fastened mates. Each mate has an offset, measured from the origin of the bottom half. The top half is raised 12 mm so that it sits on the rim of the bottom half. The Arduino block is raised 7 mm, which puts it on the floor of the bottom half. Each cap is raised 21 mm and moved sideways to its hole position from Table 4, so the flat disc sits just under the 2 mm roof and the post sticks out 6 mm above it.

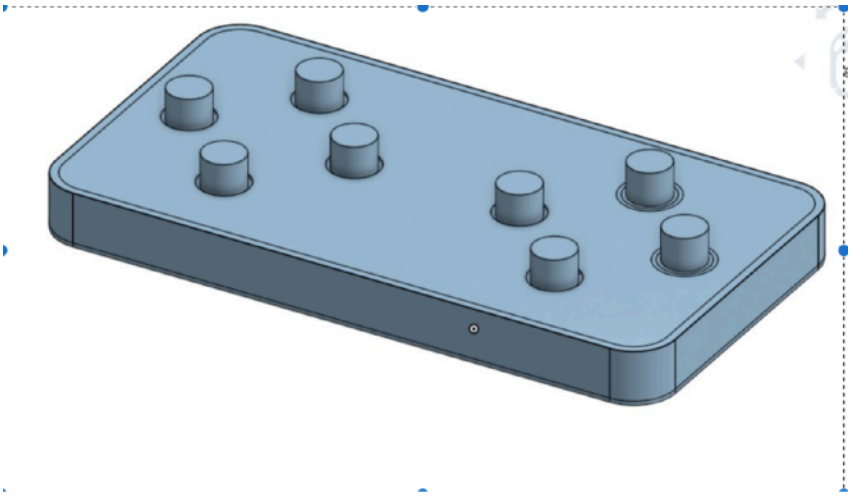


Figure 5. Assembled controller with the eight button caps

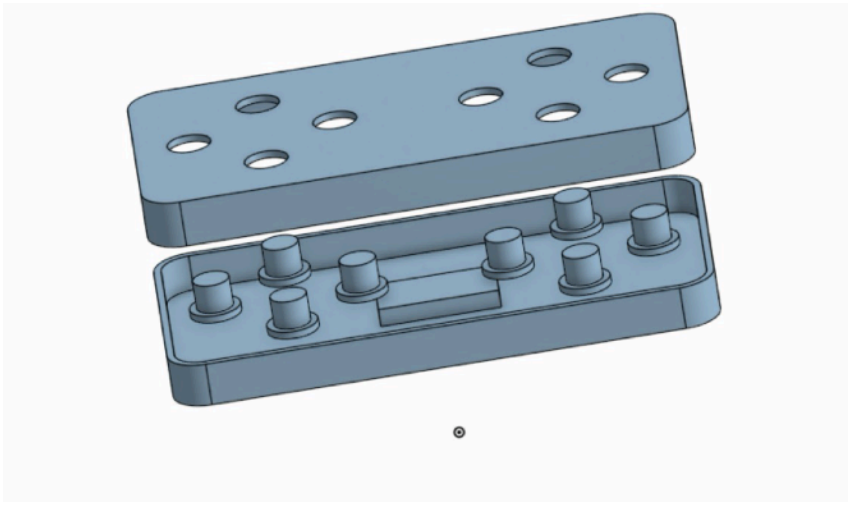


Figure 6. Exploded view: top half lifted to show the caps and the Arduino block

4.1 Mechanical checks

The fit of the parts was also checked by calculation, using the sizes in Table 3 and the hole positions in Table 4. All values in Table 5 are in millimetres.

Table 5. Mechanical checks

Check	Calculation	Result
Room around a post in its hole	(12 - 10) / 2	1 mm on each side
Flange overlap with the hole edge	(14 - 12) / 2	1 mm on each side, so the cap cannot fall out
Post height above the roof	8 - 2	6 mm
Roof material between two holes in a group	28.3 - 12	16.3 mm
Free space between two flanges in a group	28.3 - 14	14.3 mm
Hole edge to the outside of the case, Left and A	75 - (60 + 6)	9 mm
Space between the Right and Y holes	(20 - 6) - (-20 + 6)	28 mm, which holds the 18 mm Arduino block with 5 mm on each side

All checks give positive values, so no two parts collide and no hole comes too close to the edge of the case.

5. Results and verification

5.1 Electrical rules check

The KiCad Electrical Rules Checker was run on the finished schematic. It reported 0 errors and 0 warnings, which means that no wire ends in the air and the ground connection is supplied (Figure 7).

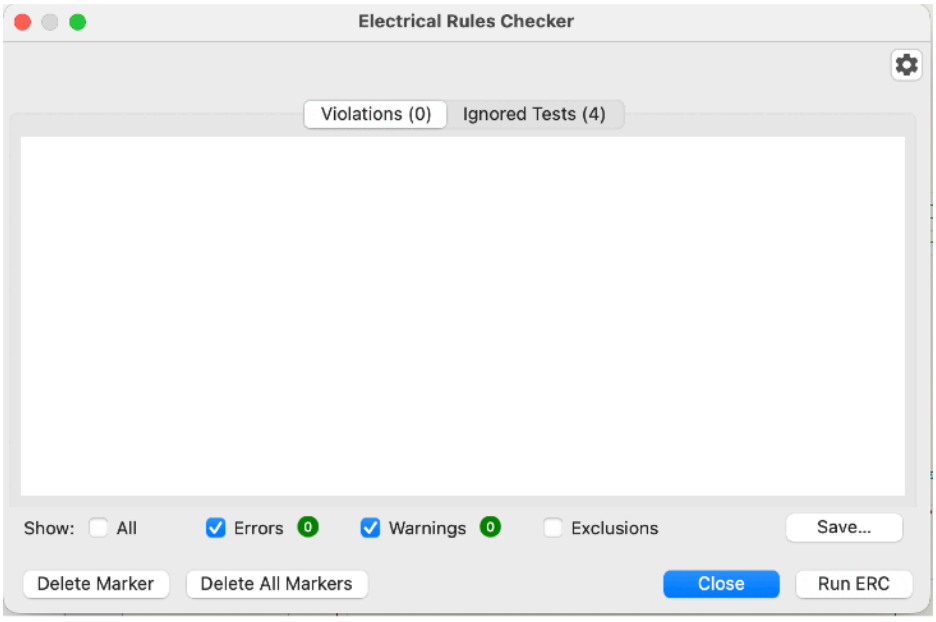


Figure 7. Result of the electrical rules check

5.2 Parts list

The parts table generated by KiCad lists nine electronic parts: eight push buttons (SW1 to SW8, named Up, Down, Left, Right, A, B, X, Y) and one Arduino Pro Micro (U1), as shown in Figure 8. This matches Table 1.

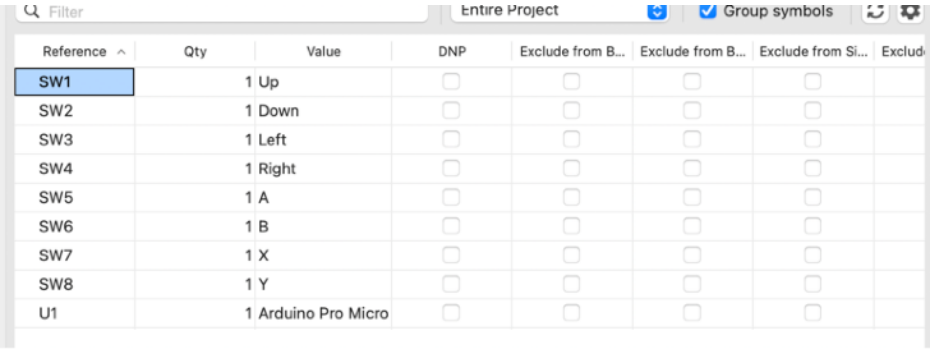


Figure 8. Parts table of the schematic

5.3 Fit of the mechanical parts

The Onshape assembly contains eleven parts joined by ten fastened mates. In the assembled view (Figure 5) all eight caps stand in their holes, and the posts stick out of the roof. In the exploded view (Figure 6) the eight flat discs sit under the roof, and the Arduino block sits on the floor between the two button groups without touching any cap. The 12 mm holes leave 1 mm of clearance around each 10 mm post.

5.4 Cost estimate

Table 6 gives a cost estimate for building one controller. The prices come from Indian seller listings seen on 25 October 2026, so they are only a guide, and a local shop may charge more for single pieces. The weight of the printed parts (about 90 g) was estimated from the volumes in the Onshape model. If the case is printed at a print shop, add its printing charge.

Table 6. Cost estimate (in Indian rupees)

Part	Quantity	Price each (₹)	Total (₹)
Pro Micro board (ATmega32U4, clone)	1	410	410

Part	Quantity	Price each (₹)	Total (₹)
12 mm by 12 mm tactile push buttons	8	7	56
Micro-USB cable, 1 m	1	129	129
3D-printed case and caps (about 90 g of PLA at ₹899 per kg)	1 set	81	81
Total			676

Sources: Pro Micro listing on IndiaMART, tactile switch listing on TradeIndia, micro-USB cable on Reliance Digital, and PLA filament on IndiaMART. PLA listings go up to about ₹2,000 per kg, which would raise the printing line to about ₹180.

5.5 Circuit simulation

To check the pull-up behaviour, a simplified model of one button channel was simulated in the Falstad circuit simulator. A 5 V source feeds a 10 kΩ resistor that stands in for the pull-up resistor, and a switch represents the push button. With the switch open (button released) the pin voltage is 5 V, which is logic HIGH (Figure 9). With the switch closed (button pressed) the pin voltage is 0 V, which is logic LOW, and a current of 500 µA flows (Figure 10). This matches Ohm's law, I = 5 V / 10 kΩ = 0.5 mA. The pull-up resistor inside the Arduino is larger (about 20 kΩ to 50 kΩ), so its real current is smaller, 0.10 mA to 0.25 mA, as calculated in section 3.2. The simulation covers only this simplified model and does not include the Arduino chip or the firmware.

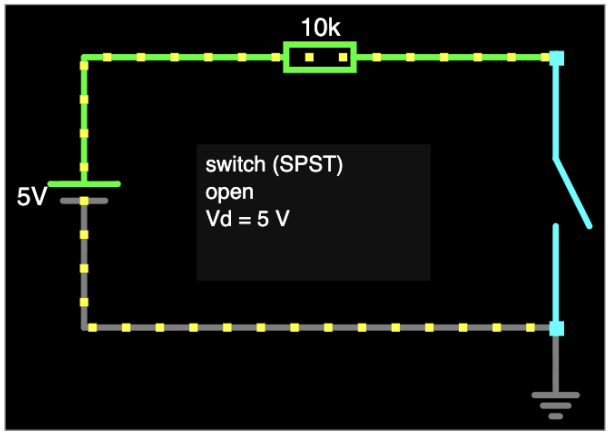


Figure 9. Button released (switch open): the pin reads 5 V, logic HIGH.

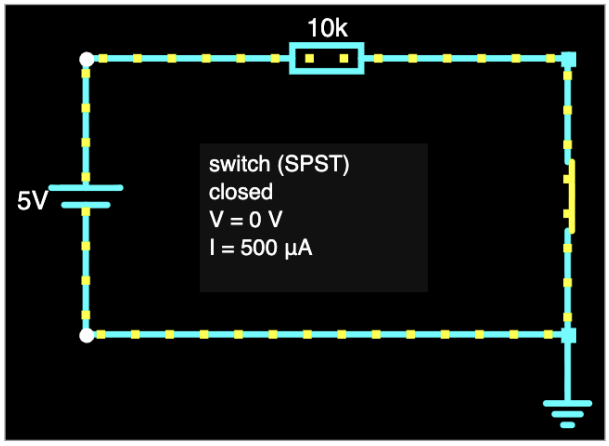


Figure 10. Button pressed (switch closed): the pin reads 0 V, logic LOW, with 500 µA flowing.

5.6 Limits of this work

This report covers the design only. The controller has not been built, and the firmware sketch in section 3.1 has not been compiled or tested on a board. The simulation in section 5.5 covers only a simplified model of one button channel, so the Arduino chip and the firmware were not simulated. The other results above come from the design checks in KiCad and from the fit in the Onshape assembly.

6. Conclusion and future work

An eight-button USB game controller was designed completely on the computer. The circuit was drawn in KiCad and passed the electrical rules check with 0 errors and 0 warnings. A simplified simulation of one button channel confirmed that the pin reads 5 V (HIGH) when the button is released and 0 V (LOW) when it is pressed. The case, the button caps and the board block were modelled and assembled in Onshape, and the caps fit their holes. The design needs only nine electronic parts and no external resistors, which keeps it cheap and easy to build.

The design can be improved in these ways:

- compile and test the firmware sketch from section 3.1 on a real board,
- add an analog joystick and shoulder buttons,
- draw a printed circuit board layout in KiCad, so that the buttons and the Arduino can sit on one board,
- add mounting posts inside the case to hold the board,
- simulate and test the Arduino and the firmware together with the buttons,
- 3D-print the case and the caps and build a working controller, and
- make a wireless version with a Bluetooth module.

7. References

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